

Group Project Task 3 – Running spark job on GCP Dataproc

Due Date: December 2, 2025, 11:59 p.m.

- Create a GCP Dataproc cluster as demonstrated in week-5 class. Refer to the slides from week-5 class.
https://usfca.instructure.com/courses/1628229/files/74534937?module_item_id=18806740
- Finalize the answer to the questions you posted in phase-1 and continued working on them in phase-2 that will create meaningful insights or data points from your dataset/s.
- Finish testing your transformations on the smaller dataset on local machine before running them on cluster.
- Create a standalone PySpark file that contains all the computations (minus the visualizations). This will be similar to the PA-2 solution file:
https://usfca.instructure.com/courses/1628229/files/74535217?module_item_id=18806739
- Submit the Dataproc batch job that will run this PySpark code successfully.
- This PySpark code should run successfully with the data that is uploaded to a Google Cloud Storage bucket.

Deliverables (as a group) **This is not an individual assignment:**

1. Screenshot of your GCP Dataproc cluster being created successfully [10 points].
2. Screenshot of the successful batch job run with success and the log outputs [20 points].
3. Submit the Python PySpark file that you uploaded to GCP bucket and ran as the batch job [20 points].
4. Upload the individual Jupyter notebooks [10 points]. This is the Jupyter notebook of your local run with the smaller dataset. Each member has to upload their own individual Jupyter notebooks. So, if there are 5 members in the team, there should be 5 notebooks.
5. Upload an exported copy in the form of html page that contains the outputs from your transformations and the visualization [10 points]. Each member must upload their own individual exports. So, if there are 5 members in the team, there should be 5 exports.

*******Submit everything as a zip file and as a group.**